Cognitive Robotics Lab 2

Piyush R Saini (23BRS1150)

Videos: https://github.com/findpiyush/cognitive-robotics

Using a python script to move the robot

Code:

```
#!/usr/bin/env python3


import sys

import select

import termios

import tty


import rclpy

from rclpy.node import Node

from geometry_msgs.msg import Twist


class KeyboardTeleop(Node):


    def __init__(self):

        super().__init__('keyboard_teleop')


        self.publisher = self.create_publisher(

            Twist,
```

```
            '/cmd_vel',
            10
        )

        # Speeds
        self.linear_speed = 0.20
        self.angular_speed = 0.80

        # Publish continuously
        self.timer = self.create_timer(
            0.1,
            self.publish_command
        )

        self.current_command = Twist()

        self.get_logger().info('Keyboard teleop started')

    def publish_command(self):
        self.publisher.publish(self.current_command)

    def set_command(self, linear, angular):
        self.current_command.linear.x = linear
        self.current_command.linear.y = 0.0
        self.current_command.linear.z = 0.0

        self.current_command.angular.x = 0.0
```

```python
        self.current_command.angular.y = 0.0
        self.current_command.angular.z = angular

    def stop(self):
        self.set_command(0.0, 0.0)

def get_key(settings):
    tty.setraw(sys.stdin.fileno())

    key = sys.stdin.read(1)

    termios.tcsetattr(
        sys.stdin,
        termios.TCSADRAIN,
        settings
    )

    return key

def main(args=None):

    rclpy.init(args=args)

    node = KeyboardTeleop()

    settings = termios.tcgetattr(sys.stdin)
```

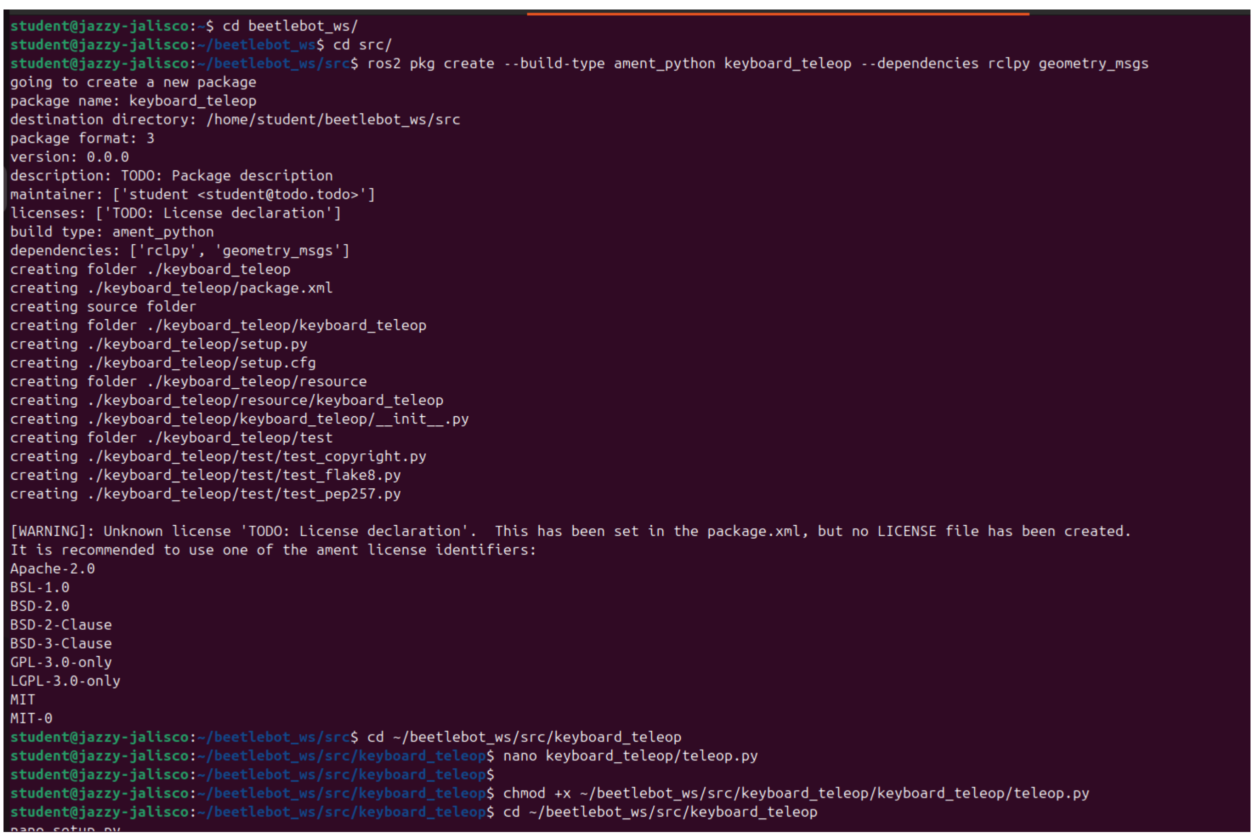
```
print('')
print('======================================')
print('       BeetleBot Keyboard Teleop')
print('======================================')
print('')
print('       w')
print('       ↑')
print('   a ← s → d')
print('       ↓')
print('       x')
print('')
print('w : Forward')
print('x : Backward')
print('a : Turn left')
print('d : Turn right')
print('s : STOP')
print('q : Quit')
print('')
print('Speed:', node.linear_speed)
print('======================================')
print('')

try:

    while rclpy.ok():

        key = get_key(settings)
```

```python
if key == 'w':
    node.set_command(
        node.linear_speed,
        0.0
    )

elif key == 'x':
    node.set_command(
        -node.linear_speed,
        0.0
    )

elif key == 'a':
    node.set_command(
        0.0,
        node.angular_speed
    )

elif key == 'd':
    node.set_command(
        0.0,
        -node.angular_speed
    )

elif key == 's':
    node.stop()
```

```
            elif key == 'q':
                node.stop()
                break

            # Let ROS process timers
            rclpy.spin_once(
                node,
                timeout_sec=0.01
            )

    except KeyboardInterrupt:
        pass

    finally:

        # Always stop the robot
        node.stop()

        # Publish several stop messages
        for _ in range(5):
            rclpy.spin_once(
                node,
                timeout_sec=0.02
            )

        termios.tcsetattr(
```

```
                sys.stdin,
                termios.TCSADRAIN,
                settings
            )

            node.destroy_node()
            rclpy.shutdown()


if __name__ == '__main__':
    main()
```

Output Screenshots:



```
student@jazzy-jalisco:~/beetlebot_ws/src$ cd ~/beetlebot_ws/src/keyboard_teleop
student@jazzy-jalisco:~/beetlebot_ws/src/keyboard_teleop$ nano keyboard_teleop/teleop.py
student@jazzy-jalisco:~/beetlebot_ws/src/keyboard_teleop$
student@jazzy-jalisco:~/beetlebot_ws/src/keyboard_teleop$ chmod +x ~/beetlebot_ws/src/keyboard_teleop/keyboard_teleop/teleop.py
student@jazzy-jalisco:~/beetlebot_ws/src/keyboard_teleop$ cd ~/beetlebot_ws/src/keyboard_teleop
nano setup.py
student@jazzy-jalisco:~/beetlebot_ws/src/keyboard_teleop$ cd ..
student@jazzy-jalisco:~/beetlebot_ws/src$ cd ..
student@jazzy-jalisco:~/beetlebot_ws$ colcon build --packages-select keyboard_teleop
Starting >>> keyboard_teleop
Finished <<< keyboard_teleop [0.81s]

Summary: 1 package finished [0.90s]
student@jazzy-jalisco:~/beetlebot_ws$ source ~/beetlebot_ws/install/setup.bash
student@jazzy-jalisco:~/beetlebot_ws$ ros2 pkg list | grep keyboard_teleop
keyboard_teleop
student@jazzy-jalisco:~/beetlebot_ws$ ros2 pkg executables keyboard_teleop
keyboard_teleop teleop
student@jazzy-jalisco:~/beetlebot_ws$
```